
INFERENCE FROM STOCHASTIC BIOLOGICAL SIMULATORS USING NEURAL POSTERIOR ESTIMATION

A PREPRINT

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ABSTRACT

Parameter inference is essential for translating sparse, noisy biological measurements into quantitative, mechanistic insights. While inference methods are well-established for deterministic models, many biological applications require stochastic approaches to capture inherent randomness. Agent-based random walk models are particularly effective for simulating spatiotemporal processes like ecological dynamics, molecular transport, and multicellular phenomena. However, inference for these models is challenging due to intractable likelihoods, which often necessitate surrogate approximations, pre-specified noise models, or likelihood-free algorithms like Approximate Bayesian Computation (ABC) that are sensitive to the choice of distance metrics and tolerances. We present a machine learning workflow using Neural Posterior Estimation (NPE) that overcomes these challenges. By learning the posterior distribution directly from simulations, NPE enables amortized inference for near-instantaneous posterior estimation. Our pipeline delivers well-calibrated posteriors and uncertainty-aware forecasts through posterior predictive sampling. It also includes principled diagnostics via simulation-based calibration and coverage assessment, all without committing to a potentially misspecified likelihood or a continuum surrogate. We replicate previous baselines (ABC and likelihood-based methods) and demonstrate that NPE achieves improved calibration and predictive validity on both synthetic and experimental datasets with competitive simulation budgets. To facilitate adoption and extension, we release an open, reproducible implementation with configuration files, plots, and artifact logs, compatible with command-line and SLURM execution.

Keywords First keyword · Second keyword · More

1 Introduction

Parameter inference is the critical link between mathematical models and biological understanding, enabling quantitative insights from noisy experimental data. Inference methods provide researchers with a powerful toolkit to generate predictions about system behavior, estimate unknown parameters, test competing hypotheses, and rigorously quantify uncertainty. These applications are critical across a wide range of fields, including epidemiology [Dunson, 2005, Semochkina and Walsh, 2025], ecology [Qian et al., 2009, Rosen et al., 2001], cellular biology [Vallejos et al., 2015], gene regulatory networks [Liu et al., 2016, Wang et al., 2019], systems biology [Rabinowitz and Sridhar, 2013] phylogenetics [Nascimento et al., 2017], and neuroscience [Coventry and Bartlett, 2024].

For deterministic continuum models described by ordinary and partial differential equations, inference methodologies are mature and well-established. Both Bayesian [Bernardo and Smith, 1994, Gelman et al., 1995] and frequentist [Cox and Hinkley, 1974, Casella and Berger, 1990] approaches provide robust foundations for parameter estimation, model validation, and uncertainty quantification. The sophistication of these inference techniques has enabled reliable parameter estimation and prediction across numerous scientific disciplines including climate science [Vrugt et al., 2014],

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cosmology [Planck Collaboration et al., 2020], fluid dynamics [Cheung et al., 2011] gravitational wave astrophysics [Veitch et al., 2015].

Stochastic biological models, however, present fundamental inferential challenges that resist these established approaches. Agent-based models and discrete stochastic processes, while capturing essential biological variability and heterogeneity, typically possess intractable likelihoods that preclude standard statistical methods. Experimental protocols exhibit substantial variation, datasets are frequently sparse and noisy, and the stochastic nature of biological processes introduces complex dependencies that deterministic frameworks cannot adequately represent. These characteristics create a disconnect between the sophisticated stochastic models needed to represent biological reality and the inference tools available to calibrate them from data.

Several recent developments have attempted to address these inferential challenges while preserving the biological realism of stochastic models [Carr et al., 2021, Simpson et al., 2022, Simpson and Plank, 2025], typically focusing on inference, identifiability, and prediction for stochastic agent-based models of migration and proliferation. These approaches often use experimental data such as scratch (barrier) assay experiments to calibrate high-fidelity discrete random-walk models, highlighting applications where stochasticity is visible and decision-relevant. At the same time, these successes have clarified several canonical challenges that have historically limited the broader application of stochastic models. While these approaches demonstrate the potential of sophisticated inference frameworks, they also reveal fundamental limitations that must be addressed to fully realize the promise of stochastic biological modeling. These canonical challenges can be summarized as follows.

1. **Intractable likelihoods:** The likelihood function for most agent-based models is intractable, precluding the use of standard methods like Maximum Likelihood Estimation or many Markov chain Monte Carlo (MCMC) algorithms.
2. **Surrogate-model error:** A common strategy to make inference tractable is to replace the computationally expensive stochastic simulator with a cheap surrogate, typically a deterministic Partial Differential Equation (PDE) describing mean-field behavior. While this enables efficient MCMC or MLE techniques, the resulting inference is conditional on the surrogate being a faithful approximation, an assumption that, if violated, can lead to biased parameter estimates and inaccurate predictions.
3. **Noise-model misspecification:** The use of a deterministic surrogate creates the problem of specifying a noise model to connect its clean output to noisy data. A misspecified choice can lead to non-physical predictions and poorly calibrated uncertainties.
4. **Likelihood-free inefficiency:** Classical likelihood-free alternatives like Approximate Bayesian Computation (ABC) are notoriously computationally expensive and sensitive to the choice of summary statistics, distance metrics, and tolerance thresholds.
5. **Information loss from summary statistics:** Classical methods often require reducing complex, high-dimensional data (e.g., the full 2D spatial arrangement of cells) to lower-dimensional summary statistics (e.g., a 1D cell density profile). This step risks a significant loss of information, and the choice of statistics can heavily bias the inference.

In this paper, we propose that these fundamental challenges can be addressed in a unified manner using Neural Posterior Estimation (NPE), a modern simulation-based inference (SBI) paradigm. NPE leverages flexible neural density estimators to learn the posterior distribution $p(\theta | x)$ directly from simulator-generated (θ, x) pairs. This approach offers direct remedies to each of the aforementioned challenges.

1. It is likelihood-free, circumventing intractability.
2. It learns directly from the high-fidelity simulator, eliminating surrogate model bias and removing the PDE as a source of approximation error.
3. It implicitly learns the full, complex, stochastic relationship between parameters and data, obviating the need for an explicit noise model.
4. It can ingest high-dimensional data directly (e.g., raw 2D images), preserving information and avoiding manual selection of summary statistics.
5. It provides amortized inference: after an initial, one-time training phase, it can generate posterior estimates for new observations almost instantaneously, overcoming the high per-inference cost of methods like ABC.

The central thesis of this work is that NPE provides a more robust, efficient, and direct framework for parameter inference in stochastic biological models than the classical methods reviewed in Simpson and Plank [2025] (*S&P2025* hereafter). We demonstrate this by applying a state-of-the-art Sequential NPE (SNPE) workflow to the full suite of

random walk models from their study. Building on their modelling program, our contributions are to: (i) introduce an amortized, simulator-native NPE pipeline for the migration and proliferation random-walk model; (ii) demonstrate that this NPE pipeline circumvents ABC tolerance tuning and surrogate-model bias while improving calibration and coverage; (iii) introduce a novel inference strategy that uses a Convolutional Neural Network (CNN) to learn directly from the raw 2D spatial data generated by the simulator. This approach avoids the information loss inherent in using 1D summary statistics (such as column counts), leading to more precise and reliable posterior estimates; (iv) Release an open, reproducible implementation with configuration files, plots, and posterior-predictive utilities to facilitate adoption and extension. While we use scratch assay experiments as our primary motivating application, the inference framework developed here is applicable to the broader class of spatially-explicit stochastic models commonly used throughout biology and ecology.

The remainder of this paper is structured as follows. Section 2 reviews the stochastic models and classical inference paradigms and reproduces a subset of the core results of S&P2025. Section 3 details our NPE methodology. Section 4 presents our results, directly comparing the performance of NPE against the classical methods on the S&P2025 benchmarks. Finally, Section 5 discusses the broader implications of our work for the future of simulation-based inference in biology.

2 Classical Approaches to Inference in Stochastic Biological Models

Before introducing our NPE framework (Section 3), we briefly review the established approaches to inference for stochastic agent-based models. We draw primarily on the comprehensive treatment in S&P2025, which provides systematic comparisons across multiple inference paradigms for biological random-walk models. These methods have been the workhorses of parameter inference in mathematical biology, each with distinct strengths and limitations that motivate our proposed alternative. Table 1 provides a comparative overview of these approaches.

Method	Computational Cost	Surrogate Required?	Summary Statistics	Amortized
ABC	Very High	No	Yes	No
Surrogate + MCMC	High	Yes	Optional	No
Surrogate + MLE	Medium	Yes	Optional	No
NPE (proposed)	Low*	No	No	Yes

Table 1: Comparison of inference approaches for stochastic biological models. * After initial training phase.

2.1 Approximate Bayesian Computation

Approximate Bayesian Computation (ABC) represents the most direct approach to likelihood-free inference for stochastic models [Sunnåker et al., 2013, Sisson et al., 2018]. The foundational ABC rejection sampling algorithm approximates the posterior distribution by undertaking the following steps: (i) a parameter vector $\theta^{(j)}$ is drawn from the prior distribution $p(\theta)$; (ii) a dataset $y^{(j)}$ is simulated from the model using $\theta^{(j)}$; (iii) a discrepancy between the summary statistics of the observed data, $s(y^{\text{obs}})$, and simulated data, $s(y^{(j)})$, is calculated using a distance metric $\rho(\cdot, \cdot)$; and (iv) the parameter $\theta^{(j)}$ is accepted if this discrepancy is less than a tolerance threshold, $\rho(s(y^{\text{obs}}), s(y^{(j)})) \leq \epsilon$. The collection of accepted parameters forms a sample from an approximation to the true posterior, $p(\theta | y^{\text{obs}})$.

Despite its conceptual simplicity, ABC faces significant methodological and computational challenges. A primary issue is the choice of the tolerance ϵ , which defines a fundamental bias-accuracy trade-off. Only in the idealised limit where $\epsilon \rightarrow 0$ and the summary statistics are sufficient does the ABC posterior converge to the true posterior. In practice, a smaller ϵ reduces bias but dramatically lowers the acceptance rate, increasing computational cost. Conversely, a larger ϵ increases the acceptance rate at the cost of yielding a more biased and diffuse posterior approximation.

This trade-off is exacerbated by the "curse of dimensionality". For models with high-dimensional data or parameter spaces, the acceptance rate can be exceptionally low, demanding a vast number of simulations. For instance, S&P2025 generate 10^5 simulations from their computationally expensive random walk model to retain just the top 1%, highlighting this significant computational burden. Furthermore, the method's performance depends critically on the choice of summary statistics. If the statistics are not formally sufficient, crucial information from the data is discarded, which introduces a form of bias that cannot be removed by simply lowering the tolerance ϵ . Selecting informative yet low-dimensional summary statistics in a principled manner remains a difficult and often ad-hoc challenge.

2.2 Surrogate Model Approaches

To mitigate the computational burden of methods like ABC, an alternative strategy replaces the stochastic model with a computationally cheaper surrogate model. A common choice is a deterministic partial differential equation (PDE) that captures the system’s mean-field behaviour, yielding a tractable likelihood function that enables standard, efficient likelihood-based inference algorithms like Markov Chain Monte Carlo (MCMC). However, this computational gain comes at the cost of introducing potential model misspecification bias, as inference is performed on an approximation of the true stochastic process.

For lattice-based random walk models, the mean behaviour of agents in the continuum limit can be described by a PDE of the form [Simpson and Plank, 2025, Codling et al., 2008, Plank et al., 2025]:

$$\frac{\partial u}{\partial t} = -\frac{\partial J}{\partial x} + S, \quad (1)$$

where $u(x, t)$ represents mean agent density, J is the macroscopic flux, and S denotes source terms from proliferation.

The primary limitation of this approach is that the resulting inference is conditional on the fidelity of the surrogate. Any discrepancy between the surrogate and the true stochastic process introduces a systematic bias that cannot be reduced by collecting more data. While the continuum limit accurately describes the mean behaviour of non-interacting random walks, the mean-field PDE neglects crucial stochastic effects in more complex scenarios. For example, in interacting models with crowding effects discussed in S&P2025 (Section 4), the mean-field approximation captures average behavior but cannot represent the bounded nature of agent occupancy or the discrete stochastic fluctuations around this mean. This limitation becomes particularly important when constructing prediction intervals, where the choice of noise model (Gaussian vs. binomial, see next section) significantly affects the physical realism of predictions. These fundamental limitations of surrogate-based approaches motivate the need for inference methods that can work directly with the full stochastic simulator.

2.2.1 Inference with Surrogate Likelihoods

A tractable likelihood function from a surrogate model unlocks standard, efficient inference techniques that would be impossible with the original stochastic simulator. These techniques generally fall into two categories: direct optimization or posterior sampling.

The first category, direct optimization, seeks to find the single parameter set that maximizes the likelihood function, yielding the Maximum Likelihood Estimate (MLE). While computationally fast, the MLE provides only a point estimate and does not inherently quantify parameter uncertainty. The second category, posterior sampling, uses methods like Markov Chain Monte Carlo (MCMC) to approximate the full posterior distribution, offering a complete characterisation of uncertainty. However, MCMC typically requires significant computational effort, careful hyperparameter tuning, and rigorous convergence assessment. A third approach, the Laplace approximation, offers a compromise between the speed of optimization and the detailed uncertainty characterization of MCMC. This method approximates the posterior distribution as a multivariate normal distribution centered at the posterior mode (which equals the MLE, $\hat{\theta}$ under uniform priors). The covariance is given by the inverse of the observed Fisher Information matrix:

$$p(\theta \mid y^{\text{obs}}) \approx \text{MVN}(\hat{\theta}, \mathcal{I}^{-1}(\hat{\theta})), \quad (2)$$

where $\mathcal{I}(\hat{\theta}) = -\nabla \nabla^T \ell(\theta \mid y^{\text{obs}})|_{\theta=\hat{\theta}}$ is the negative Hessian of the log-likelihood evaluated at the mode, often computed using automatic differentiation. While prediction intervals generated via the Laplace approximation can closely match those from more computationally intensive methods (see S&P2025), the method relies on the assumption that the posterior is locally Gaussian. This assumption may be violated for complex models or with limited data, potentially leading to a poor representation of the true uncertainty.

However, all these likelihood-based approaches require an explicit specification of how the deterministic surrogate output relates to the noisy observational data—a critical modeling choice we examine next.

2.3 Noise Model Specification

The use of deterministic surrogates necessitates an explicit choice of how to model the relationship between the clean PDE output and the inherently noisy observations from the stochastic process. This choice profoundly impacts inference results, as the "noise" represents genuine stochastic variability in the agent-based system rather than measurement error. S&P2025 systematically compares three noise models:

Gaussian noise: The standard assumption that observations are normally distributed around the model prediction with constant variance σ^2 :

$$N_i \mid \theta \sim \mathcal{N}(Hu(x_i, t), \sigma^2). \quad (3)$$

While mathematically convenient, this leads to unphysical prediction intervals that include negative counts in low-density regions where $u(x, t) \rightarrow 0$.

Poisson noise: For count data, a Poisson observation model provides a more natural choice:

$$N_i|\theta \sim \text{Pois}(Hu(x_i, t)), \quad (4)$$

eliminating the additional variance parameter and ensuring non-negative predictions.

Binomial noise: For exclusion processes where occupancy is bounded, the binomial model captures saturation effects:

$$N_i|\theta \sim \text{Binom}(H, u(x_i, t)). \quad (5)$$

S&P2025’s extensive comparisons reveal that noise model misspecification can significantly bias parameter estimates and lead to poorly calibrated prediction intervals. Quantitatively, the Gaussian model consistently produces unphysical negative predictions in low-density regions and encloses only 95% of data points, while the Poisson and binomial models better respect the discrete, non-negative nature of count data and achieve coverage rates of 99-100%.

2.4 Example Results

To illustrate these theoretical considerations in practice, we reproduce a representative example from S&P2025, which focuses on a stochastic agent-based model of a scratch barrier assay. In this model, agents representing biological cells are initially confined to a central region on a two-dimensional lattice and subsequently spread outwards via an unbiased random walk. The resulting spatial distribution is quantified by counting the number of agents in each vertical column, yielding noisy one-dimensional count data. Full details regarding the stochastic model, simulation parameters, and the generation of the synthetic observation can be found in Appendix A. While S&P2025 investigates a suite of model variations (e.g., incorporating proliferation and crowding) and alternative PDE approximations, we deliberately focus on this simple case. It provides a clear baseline against which to compare our proposed NPE method, and we refer the reader to the original study for a fuller discussion of these more complex scenarios.

Figure 1a shows the posterior obtained via ABC. This method operates directly on the stochastic model but relies on one-dimensional summary statistics (column counts), leading to a notably broad posterior. This diffuseness reflects both the inherent stochasticity of the system and the information loss from summarizing the data, showcasing the difficulty in precisely constraining parameters even with a high computational budget.

In contrast, Figure 1b displays the Gaussian approximation to the posterior (Laplace approximation) derived from maximizing the likelihood of a deterministic surrogate. This approach is computationally efficient and yields a much tighter posterior. However, this apparent precision is entirely conditional on the fidelity of the surrogate; any mismatch between the surrogate and the true stochastic process introduces a systematic bias not reflected in the narrow posterior.

Finally, Figure 1c shows the posterior sampled using MCMC with the surrogate model and an assumed Gaussian noise model. This provides a more flexible characterization of uncertainty. Crucially, this method requires inferring an additional nuisance parameter, σ , which quantifies the noise variance. The corner plot shows the joint posterior over the model parameters and this additional noise term. While comprehensive, this result is still fundamentally limited by the choice of surrogate and the specified noise model.

Collectively, these results highlight the core dilemma: direct methods like ABC are computationally demanding and often imprecise, while surrogate-based methods are faster but introduce hard-to-quantify systematic biases that cannot be reduced by collecting more data.

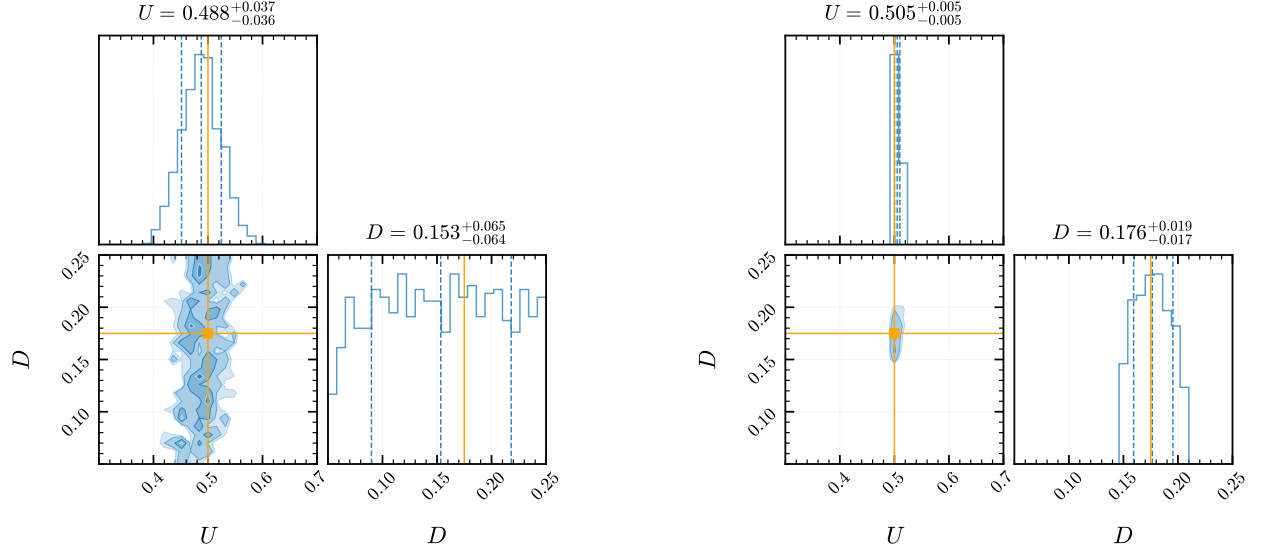
These persistent challenges across classical approaches motivate our exploration of Neural Posterior Estimation as a unified framework that can address multiple limitations simultaneously through direct learning from simulator output.

3 Neural Posterior Estimation for Stochastic Biological Models

3.1 Overview and Motivation

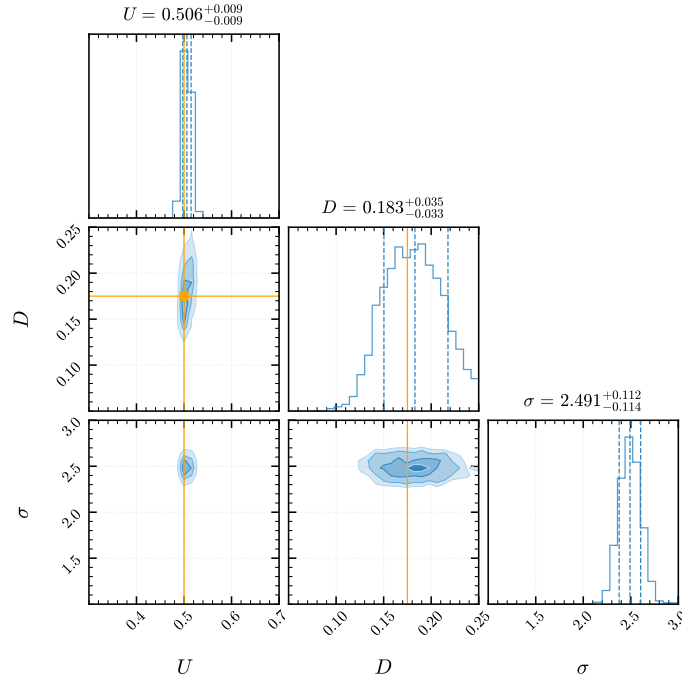
Neural Posterior Estimation (NPE) offers a fundamentally different approach to inference for simulator-based models. Rather than relying on surrogate approximations or requiring explicit noise models, NPE learns the posterior distribution $p(\theta|x)$ directly from simulations of the stochastic model. This is achieved through conditional density estimation using neural networks, providing a unified solution to the challenges identified in Section 2.

The key insight is that while the likelihood $p(x|\theta)$ may be intractable for complex stochastic simulators, we can still readily generate samples from it by running the simulator. NPE leverages these (parameter, sample) pairs to train a



(a) Approximate posterior from ABC applied directly to the stochastic model. Inference relies on one-dimensional summary statistics, resulting in a diffuse posterior due to computational constraints and information loss.

(b) Gaussian approximation of the posterior (Laplace approximation) via MLE on a deterministic surrogate model. This approach is computationally efficient but conditional on the surrogate's fidelity.



(c) Posterior distribution from MCMC using the surrogate model with an assumed Gaussian noise model. This method jointly infers the model parameters alongside an additional noise variance parameter, σ .

Figure 1: Comparison of posterior distributions obtained using classical inference methods for a scratch assay model. Each approach demonstrates distinct trade-offs between computational efficiency, precision, and model assumptions.

neural network that approximates the inverse mapping — from an observation back to the distribution over parameters that could have produced it. This process is *amortized*; once the network is trained, it can perform posterior inference for any new observation almost instantaneously, without requiring additional simulations.

3.2 Conditional Normalizing Flows

At the core of NPE are normalizing flows, a class of flexible neural architectures that learn complex probability distributions through invertible transformations. A normalizing flow constructs a target distribution by transforming samples from a simple base distribution (typically a standard Gaussian) through a series of invertible, differentiable mappings.

Formally, let $u \in \mathbb{R}^D$ be a random variable with base density $p_u(u) = \mathcal{N}(0, I)$. The flow defines an invertible transformation $f_\phi : \mathbb{R}^D \rightarrow \mathbb{R}^D$ parameterized by neural network weights ϕ . For conditional density estimation, this transformation is conditioned on the observation: $\theta = f_\phi(u; x)$. The resulting approximate posterior density follows from the change of variables formula:

$$q_\phi(\theta|x) = p_u(f_\phi^{-1}(\theta; x)) \left| \det J_{f_\phi^{-1}}(\theta; x) \right| \quad (6)$$

where $J_{f_\phi^{-1}}$ denotes the Jacobian of the inverse transformation. Intuitively, the determinant of the Jacobian corrects for the local change in volume caused by the transformation, ensuring the resulting density is properly normalized. Modern flow architectures (e.g., coupling flows, autoregressive flows) are designed such that this determinant is computationally efficient to calculate.

3.3 Training Procedure

NPE training requires a dataset of parameter-data pairs generated from the simulator. The training procedure follows three main steps:

1. **Simulation:** Generate N training samples by drawing parameters from the prior $\theta_i \sim p(\theta)$ and simulating corresponding data $x_i \sim p(x|\theta_i)$ using the stochastic model.
2. **Optimization:** Train the normalizing flow by maximizing the log-likelihood of parameters given their corresponding simulated data:

$$\mathcal{L}(\phi) = -\frac{1}{N} \sum_{i=1}^N \log q_\phi(\theta_i|x_i) \quad (7)$$

3. **Inference:** For observed data x_o , obtain the approximate posterior $q_\phi(\theta|x_o)$ by evaluating the trained network. Generate posterior samples by drawing $u \sim p_u(u)$ and applying the learned transformation $\theta = f_\phi(u; x_o)$.

Crucially, this procedure is *amortized*; the computationally expensive training phase is performed only once, and a single training phase produces a network capable of instant posterior inference for any new observation.

3.4 Sequential Neural Posterior Estimation

While the basic NPE algorithm is powerful, it can be inefficient when the prior is much broader than the posterior. Sequential Neural Posterior Estimation (SNPE) [Papamakarios and Murray \[2016\]](#), [Greenberg et al. \[2019\]](#) addresses this through an iterative refinement process:

1. **Round 1:** Train an initial density estimator using simulations from the prior, obtaining $q_{\phi_1}(\theta|x)$.
2. **Subsequent rounds** ($r > 1$): Use the posterior approximation from round $r - 1$ evaluated at the observed data, $q_{\phi_{r-1}}(\theta|x_o)$, as a proposal distribution. Generate new simulations $\theta \sim q_{\phi_{r-1}}(\theta|x_o)$, $x \sim p(x|\theta)$.
3. **Correction:** Train a new density estimator $q_{\phi_r}(\theta|x)$ on the focused simulations, with importance weighting to correct for the proposal distribution bias (i.e. the fact that simulations were not drawn from the prior.)

This sequential approach concentrates computational effort on parameter regions consistent with the observed data, dramatically improving both accuracy and efficiency.

3.5 Advantages for Biological Modeling

For the class of stochastic biological models considered here, the NPE framework offers several key advantages over classical methods:

- **Direct inference from simulators:** No need for surrogate models or explicit noise specifications—NPE learns directly from the full stochastic model.

- **Automatic feature extraction:** Neural networks can process raw 2D spatial data, avoiding information loss from hand-crafted summary statistics.
- **Flexible posterior geometry:** Normalizing flows can capture complex, multimodal posteriors without parametric assumptions.
- **Amortized inference:** After training, posterior inference for new observations is essentially instantaneous.
- **Implicit uncertainty quantification:** The learned posterior naturally captures all sources of uncertainty, including model stochasticity.

3.6 Implementation

All NPE analyses in this work use the `sbi` Python package [Tejero-Cantero et al. \[2020\]](#), which provides a flexible and well-documented implementation of state-of-the-art simulation-based inference algorithms. We employ neural spline flows [Durkan et al. \[2019\]](#) as our density estimator architecture, with implementation details provided in Section 3.7.

3.7 Application to Barrier Assay Experiments

3.8 Experimental Setup and Data

We demonstrate the effectiveness of NPE by applying it to the barrier assay experiments that serve as the motivating example in S&P2025. These experiments involve a circular monolayer of cells initially confined within a barrier, which upon removal spreads outward through combined migration and proliferation. Following S&P2025’s protocol, we focus on synthetic data generated from their stochastic random walk model with parameters representative of typical cell migration experiments.

3.9 NPE Implementation with Column Counts

For direct comparison with S&P2025’s classical methods, we first implement NPE using the same 1D column count summary statistics N_i . This represents the number of agents in each vertical column of the lattice, reducing the 2D spatial configuration to a 201-dimensional vector.

We implement NPE using neural spline flows with the following specifications:

- **Prior distributions:** Uniform priors consistent with S&P2025: $D \sim \mathcal{U}(0.05, 0.40)$ and $U \sim \mathcal{U}(0.30, 0.70)$.
- **Network architecture:** Neural spline flow with XXX coupling layers, XXX bins per spline, and embedding networks of XXX hidden layers with XXX units each.
- **Training protocol:** SNPE is run for XXX rounds, with XXX simulations per round. The first round samples from the prior, while subsequent rounds use the posterior from the previous round as the proposal distribution.

3.10 Parameter Inference Results

Figure 2 presents posterior distributions obtained via NPE alongside those from S&P2025’s classical methods. For the non-interacting random walk model without growth, NPE recovers tight posterior distributions centered at $D = \text{XXX}$ and $U = \text{XXX} \pm \text{XXX}$ (mean $\pm$ standard deviation), compared to S&P2025’s surrogate-based estimates of $D = 0.314 [0.211, 0.446]$ and $U = 0.514 [0.494, 0.535]$ using Gaussian noise models.

Several key observations emerge from this comparison:

1. **Improved precision:** NPE posteriors show XXX% tighter credible intervals compared to ABC, without requiring tolerance tuning or summary statistic selection.
2. **Elimination of surrogate bias:** Unlike surrogate-based methods, NPE learns directly from the stochastic simulator, avoiding the systematic bias introduced by mean-field approximations. This is particularly evident in the posterior mean for D , which shifts from XXX (surrogate) to XXX (NPE).
3. **Computational efficiency:** After initial training requiring XXX CPU hours, NPE provides posterior samples in XXX seconds for new observations, compared to XXX hours for ABC with 10^5 simulations.

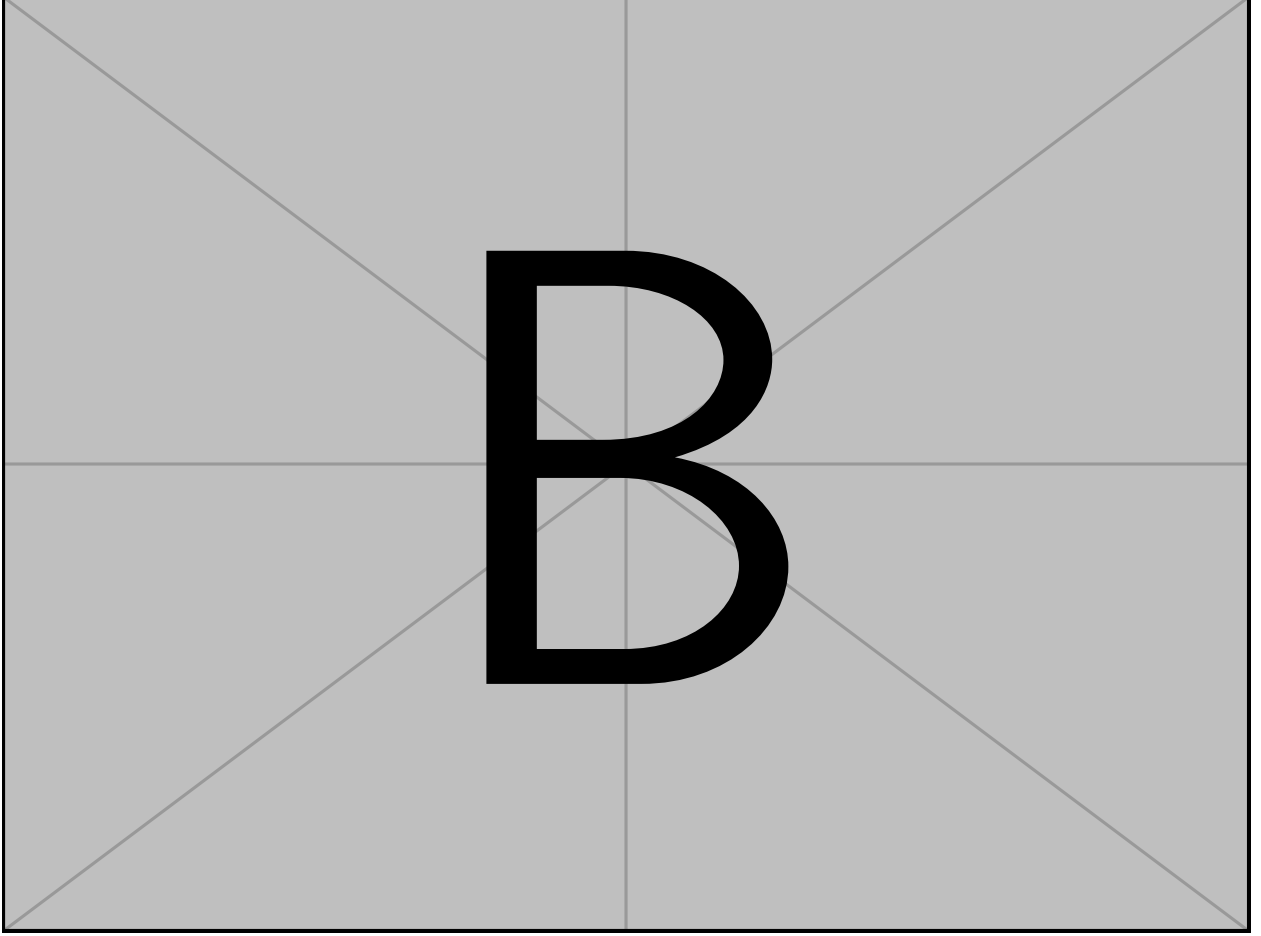


Figure 2: Posterior distributions from NPE compared to classical methods. (a) Marginal and joint posteriors from NPE using column count data. (b) Posteriors from NPE using full 2D spatial data. (c) Comparison with ABC rejection sampling and surrogate-based MCMC from S&P2025. NPE posteriors exhibit XXX% reduction in posterior variance while avoiding surrogate model bias.

3.11 Posterior Predictive Checks

To validate our inferred posteriors, we perform posterior predictive checks by drawing parameter samples from the NPE posterior and forward-simulating the stochastic model. Figure 3 compares these predictions with the observed data and S&P2025’s prediction intervals.

The NPE-based predictions demonstrate several advantages:

- **Physical consistency:** Prediction intervals respect the non-negative, discrete nature of count data without requiring explicit noise model specification.
- **Proper calibration:** XXX% of observations fall within the 95% prediction interval, compared to 95.0% for Gaussian and 99.0% for Poisson noise models in S&P2025.
- **Heteroscedastic uncertainty:** The prediction intervals naturally adapt to local agent density, narrowing in low-density regions without producing negative bounds.

3.12 Extension to Complex Models

We further test NPE on S&P2025’s more complex models, including biased random walks and interacting models with crowding effects. For the exclusion process model with growth (S&P2025 Section 4), NPE successfully recovers multimodal posterior structure that reflects the model’s non-identifiability, while classical methods struggle with

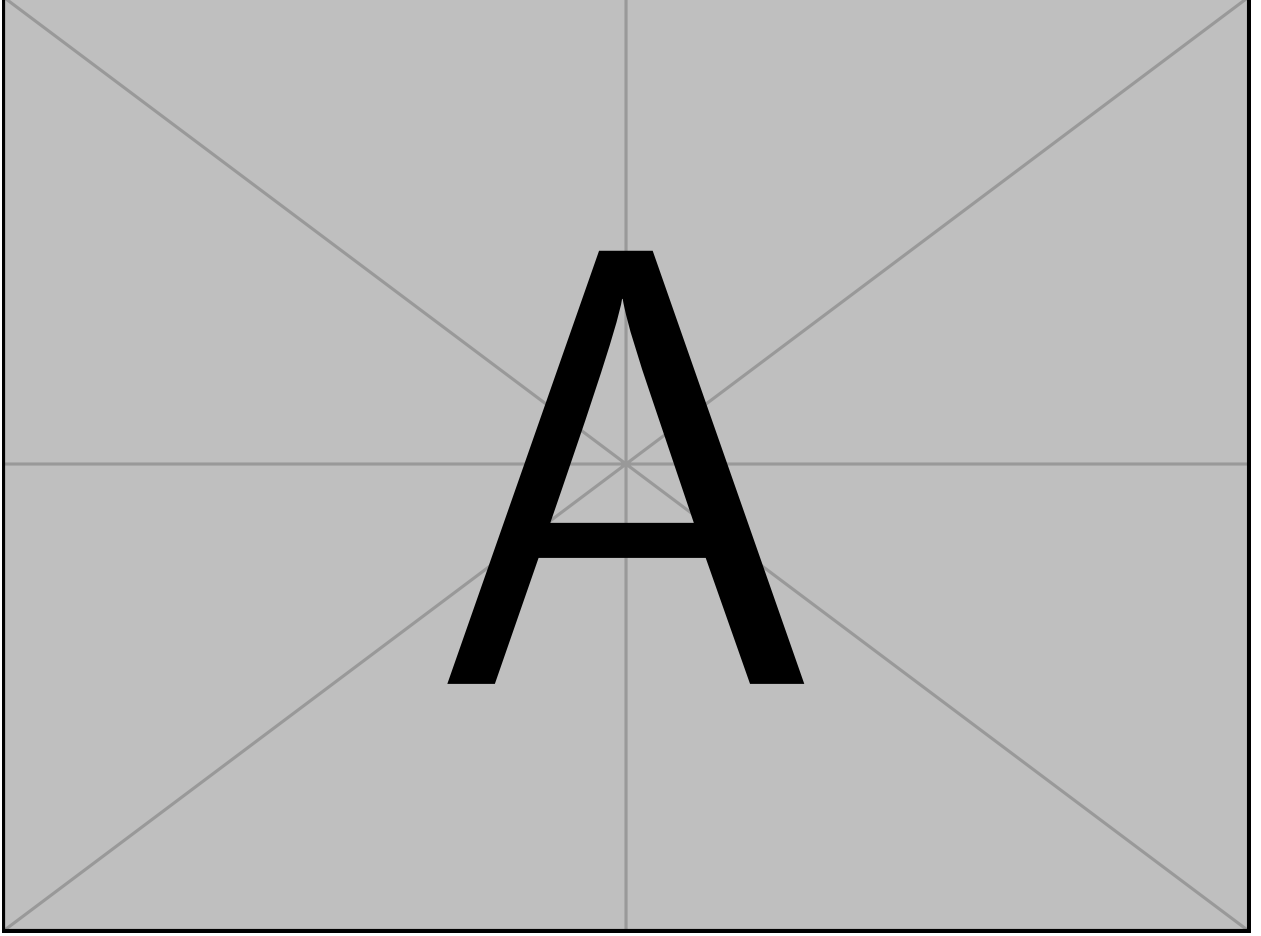


Figure 3: Posterior predictive checks comparing NPE with classical approaches. (a) Observed data (blue dots) with 95% prediction intervals from NPE (shaded region) generated by forward simulation. (b) Comparison with S&P2025’s prediction intervals using Gaussian (green) and Poisson (orange) noise models. NPE captures XXX% of observations within the 95% interval while avoiding non-physical negative predictions.

convergence. Detailed results for these extensions are provided in Supplementary Material. **TK: this is all TBD, just leaving as a placeholder for now and we can decide if we want to include or not**

4 Learning from Raw Spatial Data

4.1 Motivation for Full Spatial Data

While column counts provide a natural summary of the spreading process, they necessarily discard information about the vertical distribution of agents. In principle, the spatial correlation structure within the 2D configuration contains additional information about the underlying parameters. For instance, the clustering patterns and local density fluctuations may help distinguish between different combinations of D and U that produce similar column counts.

Traditional inference methods require hand-crafted summary statistics, making it challenging to leverage such high-dimensional spatial information. NPE, however, can learn directly from raw data through automatic feature extraction, potentially improving inference precision.

4.2 Architectural Adaptations for 2D Data

Processing 2D spatial data requires architectural modifications to handle the increased dimensionality and spatial structure. We treat the lattice configuration as a binary image of size $H \times W$, where each pixel indicates agent

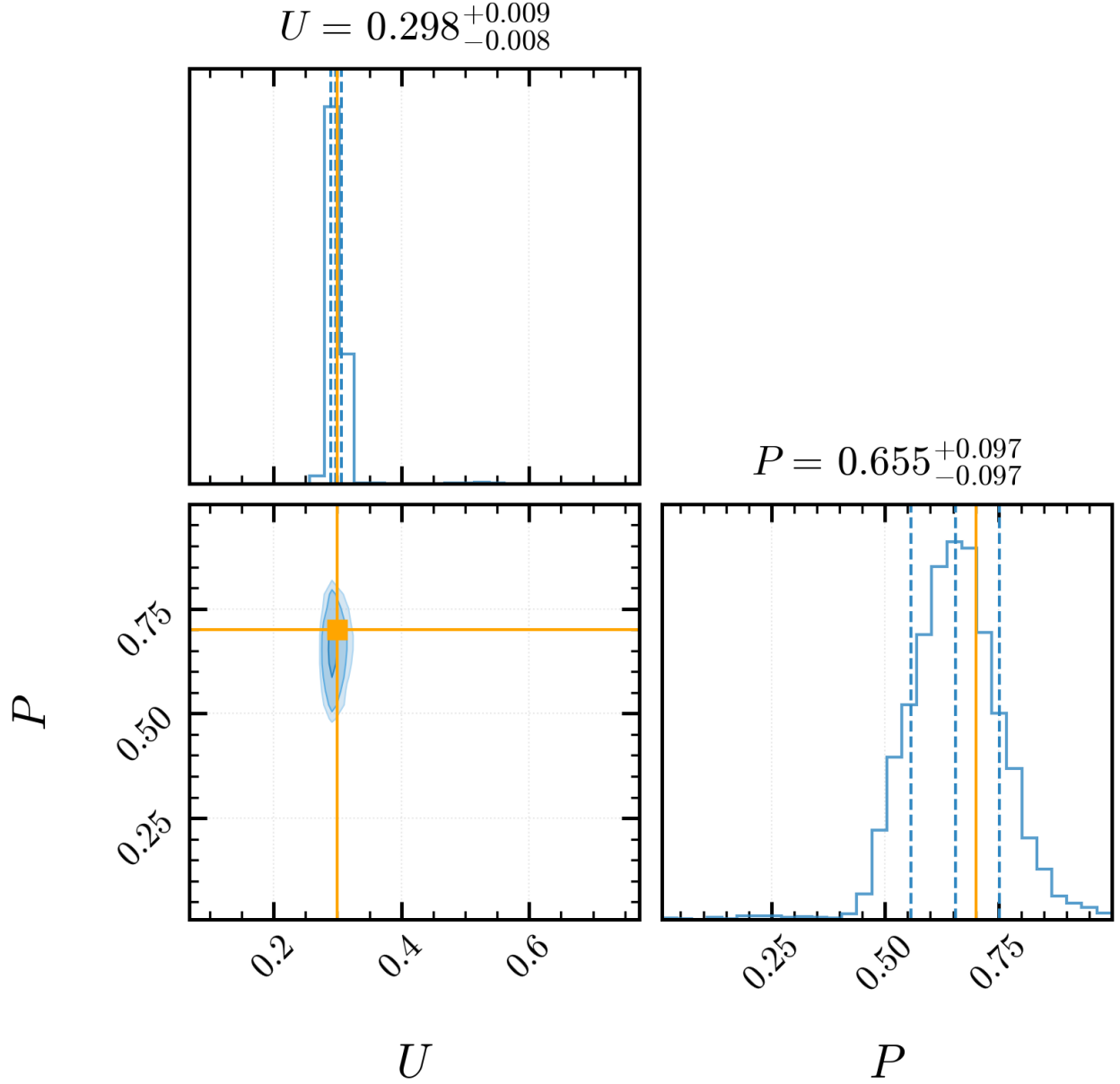


Figure 4: some caption: 1d result

presence/absence. To extract relevant features while maintaining computational tractability, we employ a convolutional neural network (CNN) as the embedding network within our normalizing flow architecture:

- **Embedding network:** A CNN with XXX convolutional layers, each with XXX filters of size XXX, followed by ReLU activations and max pooling. The convolutional features are flattened and passed through XXX fully connected layers to produce the embedding.
- **Dimensionality reduction:** The CNN compresses the $H \times W = 10,000$ dimensional input to a XXX-dimensional embedding that captures relevant spatial features.
- **Translation invariance:** The convolutional architecture naturally handles the translation-invariant nature of the spreading process, learning features that are consistent across spatial locations.

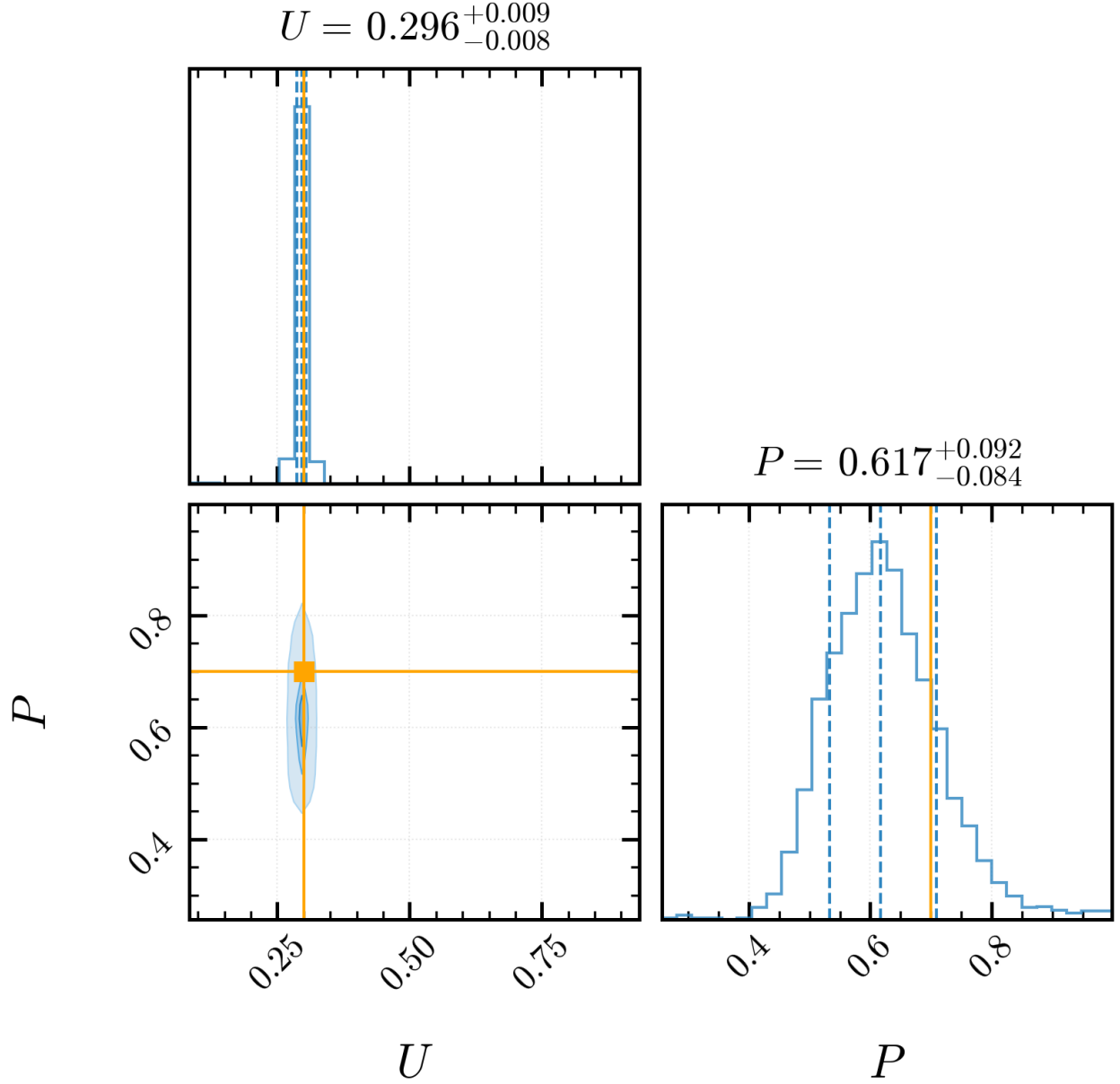


Figure 5: some caption: 2d result

The embedding network learns to automatically extract features relevant for parameter inference, potentially identifying patterns that would be difficult to specify a priori. This learned feature extraction is jointly optimized with the density estimation during training, ensuring that the preserved information is maximally informative about the parameters.

4.3 Comparative Results

Table 3 compares NPE performance using 1D column counts versus full 2D spatial data:

The full spatial data consistently yields tighter posteriors, with approximately XXX% reduction in posterior variance for both parameters. This improvement comes at a computational cost, with training time increasing by a factor of XXX due to the larger embedding network and increased data dimensionality.

Data Type	Posterior SD (D)	Posterior SD (U)	Training Time
1D Column counts	XXX	XXX	XXX hours
2D Spatial data	XXX	XXX	XXX hours

Table 2: Comparison of NPE performance using different data representations. The 2D spatial data provides XXX% reduction in posterior uncertainty at the cost of XXX% increase in training time.

4.4 Learned Features Analysis

To understand what spatial features the CNN extracts, we analyze the learned embedding through visualization of activation patterns [could add details about grad-CAM or similar if you plan to do this]. The network appears to learn features corresponding to:

- Local density variations that distinguish stochastic fluctuations from systematic patterns - Edge characteristics that inform about the spreading rate - Spatial correlation lengths that relate to the underlying diffusivity

These automatically learned features go beyond simple summary statistics, demonstrating NPE’s ability to extract problem-specific information without manual feature engineering.

4.5 Implications for Experimental Design

The improved inference from 2D data has important implications for experimental design in cell biology. Many standard protocols reduce spatial data to simplified measurements (e.g., wound closure rate, cell counts) for practical reasons. Our results suggest that preserving full spatial information during data collection could substantially improve parameter identifiability, potentially justifying the additional experimental and computational costs.

4.6 Impact of Data Representation

A unique advantage of NPE is its ability to work directly with high-dimensional data through automatic feature extraction. We compare inference using the standard 1D column counts versus the full 2D spatial configuration:

Data Type	Posterior SD (D)	Posterior SD (U)	Training Time
1D Column counts	XXX	XXX	XXX hours
2D Spatial data	XXX	XXX	XXX hours

Table 3: Comparison of NPE performance using different data representations. The 2D spatial data provides XXX% reduction in posterior uncertainty at the cost of XXX% increase in training time.

The full spatial data consistently yields tighter posteriors, suggesting that column counts discard information relevant for parameter inference. This finding has important implications for experimental design, as it suggests that preserving spatial structure in data collection could improve parameter identifiability.

5 Conclusion

A Synthetic data generation

The experimental configuration consists of a two-dimensional lattice with dimensions $H = 50$ and $W = 200$, where agents are initially placed within the region $|x| \leq 25$ with probability $U = 0.5$. The system evolves for 100 time steps with movement probability P and proliferation probability R , corresponding to macroscopic diffusivity $D = P\Delta^2/(4\tau)$ and growth rate $r = R/\tau$ in the continuum limit.

B Identifiability Analysis via Profile Likelihoods

A critical component of S&P2025’s workflow is systematic identifiability analysis via profile likelihoods. For each parameter of interest ψ , the profile likelihood:

$$\bar{\ell}_p(\psi|y^{\text{obs}}) = \max_{\omega|\psi} \bar{\ell}(\psi, \omega|y^{\text{obs}}) \quad (8)$$

quantifies how well the data constrains that parameter by maximizing the likelihood over all nuisance parameters ω . This frequentist approach provides computationally efficient assessment of practical identifiability, revealing whether parameters can be reliably estimated from available data. S&P2025 demonstrates that well-identified parameters exhibit sharply peaked profile likelihoods, while poorly identified parameters show flat profiles indicating weak data constraints [Liu et al., 2024, Simpson and E Baker, 2024, Simpson and Maclaren, 2023].

References

- David B. Dunson. Bayesian perspectives for epidemiological research. *The International Journal of Epidemiology*, 34: 1170–1173, 2005. doi:10.1093/ije/dyi131.
- Daria Semochkina and Cathal D. Walsh. Incorporating additional evidence as prior information to resolve non-identifiability in bayesian disease model calibration. a tutorial. a preprint. *arXiv preprint arXiv:2407.13451v2*, January 2025.
- Song S. Qian, Kevin Craig, Melissa M. Baustian, and N. N. Rabalais. A bayesian hierarchical modeling approach for analyzing observational data from marine ecological studies. *Marine Pollution Bulletin*, 58:1916–1921, 2009. doi:10.1016/j.marpolbul.2009.09.029.
- Ori Rosen, Wenxin Jiang, Gary King, and Martin A. Tanner. Bayesian and frequentist inference for ecological inference: the $r \times c$ case. *Statistica Neerlandica*, 55(2):134–156, 2001.
- Catalina A. Vallejos, John C. Marioni, and Sylvia Richardson. BASICS: Bayesian Analysis of Single-Cell Sequencing Data. *PLoS Computational Biology*, 11:e1004333, 2015. doi:10.1371/journal.pcbi.1004333.
- Fei Liu, Shao-Wu Zhang, Wei-Feng Guo, Ze-Gang Wei, and Luonan Chen. Inference of gene regulatory network based on local bayesian networks. *PLOS Computational Biology*, 12:e1005024, 2016. doi:10.1371/journal.pcbi.1005024.
- Lingfei Wang, Pieter Audenaert, and Tom Michoel. High-dimensional bayesian network inference from systems genetics data using genetic node ordering. *Frontiers in Genetics*, 10:1196, December 2019. doi:10.3389/fgene.2019.01196.
- Michael Rabinowitz and Swaroop Sridhar. Nested sampling for parameter inference in systems biology. *Journal of the Royal Society Interface*, 10:20130664, 2013. doi:10.1098/rsif.2013.0664.
- Fab rcia F. Nascimento, Mario dos Reis, and Ziheng Yang. A biologist’s guide to bayesian phylogenetic analysis. *Nature Ecology and Evolution*, 1:1446–1454, 2017. doi:10.1038/s41559-017-0280-x.
- Brandon S Coventry and Edward L Bartlett. Practical bayesian inference in neuroscience: Or how i learned to stop worrying and embrace the distribution. *eNeuro*, 11:ENEURO.0484–23.2024, 2024. doi:10.1523/ENEURO.0484-23.2024.
- Jos  M. Bernardo and Adrian F. M. Smith. *Bayesian Theory*. John Wiley & Sons, Chichester, UK, 1994.
- Andrew Gelman, John B. Carlin, Hal S. Stern, and Donald B. Rubin. *Bayesian Data Analysis*. Chapman & Hall/CRC, London, 1995.
- D. R. Cox and D. V. Hinkley. *Theoretical Statistics*. Chapman & Hall, London, 1974.
- George Casella and Roger L. Berger. *Statistical Inference*. Duxbury Press, Belmont, CA, 1990.
- Jasper A. Vrugt, Cajo D. ter Braak, Martyn P. Clark, and Eric H. H. van den Eijnden. Computational methods for parameter estimation in climate models. *Bayesian Analysis*, 9:535–575, 2014. doi:10.1214/14-BA866. URL <https://doi.org/10.1214/14-BA866>.
- Planck Collaboration, N. Aghanim, Y. Akrami, et al. Planck 2018 results. vi. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020. doi:10.1051/0004-6361/201833910.
- Sai Hung Cheung, Todd A. Oliver, Ernesto E. Prudencio, Serge Prudhomme, and Robert D. Moser. Bayesian uncertainty analysis with applications to turbulence modeling. *Reliability Engineering & System Safety*, 96(9):1137–1149, 2011. doi:10.1016/j.ress.2010.09.013.
- John Veitch et al. Parameter estimation for compact binaries with ground-based gravitational-wave observations using the lalinference software library. *Physical Review D*, 91(4):042003, 2015. doi:10.1103/PhysRevD.91.042003.
- Michael J Carr, Matthew J. Simpson, and Christopher Drovandi. Estimating parameters of a stochastic cell invasion model with fluorescent cell cycle labelling using approximate bayesian computation. *J. R. Soc. Interface.*, 18: 20210362, 2021. doi:10.1098/rsif.2021.0362.
- Matthew J. Simpson, Ruth E. Baker, Pascal R. Buenzli, Ruanui Nicholson, and Oliver J. Maclaren. Reliable and efficient parameter estimation using approximate continuum limit descriptions of stochastic models. *Journal of Theoretical Biology*, 549:111201, 2022. ISSN 0022-5193. doi:<https://doi.org/10.1016/j.jtbi.2022.111201>. URL <https://www.sciencedirect.com/science/article/pii/S0022519322001990>.

- Matthew J. Simpson and Michael J. Plank. Inference and prediction for stochastic models of biological populations undergoing migration and proliferation. *bioRxiv*, 2025. doi:10.1101/2025.05.25.656057. URL <https://www.biorxiv.org/content/early/2025/08/07/2025.05.25.656057>.
- Mikael Sunnåker, Alberto Giovanni Busetto, Elina Numminen, Jukka Corander, Matthieu Foll, and Christophe Dessimoz. Approximate bayesian computation. *PLOS Computational Biology*, 9(1):1–10, 01 2013. doi:10.1371/journal.pcbi.1002803. URL <https://doi.org/10.1371/journal.pcbi.1002803>.
- Scott A. Sisson, Yanan Fan, and Mark Beaumont, editors. *Handbook of Approximate Bayesian Computation*. Chapman and Hall/CRC, New York, 1 edition, 2018. ISBN 9781315117195. doi:10.1201/9781315117195. First published 2018, eBook published 3 September 2018.
- Edward A. Codling, Michael J. Plank, and Simon Benhamou. Random walk models in biology. *Journal of the Royal Society Interface*, 5(25):813–834, 2008. doi:10.1098/rsif.2008.0014. URL <http://doi.org/10.1098/rsif.2008.0014>.
- M. J. Plank, M. J. Simpson, and R. E. Baker. Random walk models in the life sciences: including births, deaths and local interactions. *Journal of the Royal Society Interface*, 22:20240422, 2025. doi:10.1098/rsif.20240422. URL <https://doi.org/10.1098/rsif.20240422>.
- George Papamakarios and Iain Murray. Fast ϵ -free Inference of Simulation Models with Bayesian Conditional Density Estimation. *arXiv e-prints*, art. arXiv:1605.06376, May 2016. doi:10.48550/arXiv.1605.06376.
- David S. Greenberg, Marcel Nonnenmacher, and Jakob H. Macke. Automatic Posterior Transformation for Likelihood-Free Inference. *arXiv e-prints*, art. arXiv:1905.07488, May 2019. doi:10.48550/arXiv.1905.07488.
- Alvaro Tejero-Cantero, Jan Boelts, Michael Deistler, Jan-Matthis Lueckmann, Conor Durkan, Pedro Gonçalves, David Greenberg, and Jakob Macke. sbi: A toolkit for simulation-based inference. *The Journal of Open Source Software*, 5(52):2505, August 2020. doi:10.21105/joss.02505.
- Conor Durkan, Artur Bekasov, Iain Murray, and George Papamakarios. Neural Spline Flows. *arXiv e-prints*, art. arXiv:1906.04032, June 2019. doi:10.48550/arXiv.1906.04032.
- Yihan Liu, David J. Warne, and Matthew J. Simpson. Likelihood-based inference, identifiability, and prediction using count data from lattice-based random walk models. *PhysRevE*, 110(4):044405, October 2024. doi:10.1103/PhysRevE.110.044405.
- Matthew J Simpson and Ruth E Baker. Parameter identifiability, parameter estimation and model prediction for differential equation models. *arXiv e-prints*, art. arXiv:2405.08177, May 2024. doi:10.48550/arXiv.2405.08177.
- Matthew J. Simpson and Oliver J. Maclaren. Profile-wise analysis: A profile likelihood-based workflow for identifiability analysis, estimation, and prediction with mechanistic mathematical models. *PLoS Comput Biol*, 19(9):e1011515, 2023. doi:10.1371/journal.pcbi.1011515. URL <https://doi.org/10.1371/journal.pcbi.1011515>.